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Security Mindset: Beyond Tools, Into Reality

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#raconf2026

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From the logical operators to RCE

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Round 1: Finding a Zero-Click Entry Point

- **Goal:** Remote compromise
- **Constraint:** No user interaction
- **Target:** Modern mobile OS (iOS)

but, where do we look?

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- Wireless Protocols (Wi-Fi, Bluetooth)
- Image Decoding (JPEG, GIF, JBIG2)
- Media Processing (Audio, Video)

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Round 2: Attack Surface

- iMessage delivery pipeline
- Auto-processed attachments
- No user interaction

- Untrusted web content
- Complex rendering engine
- Attacker-controlled input

- Media preview parsing
- Background processing
- External input

- Image decoding pipeline
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- Auto thumbnail generation

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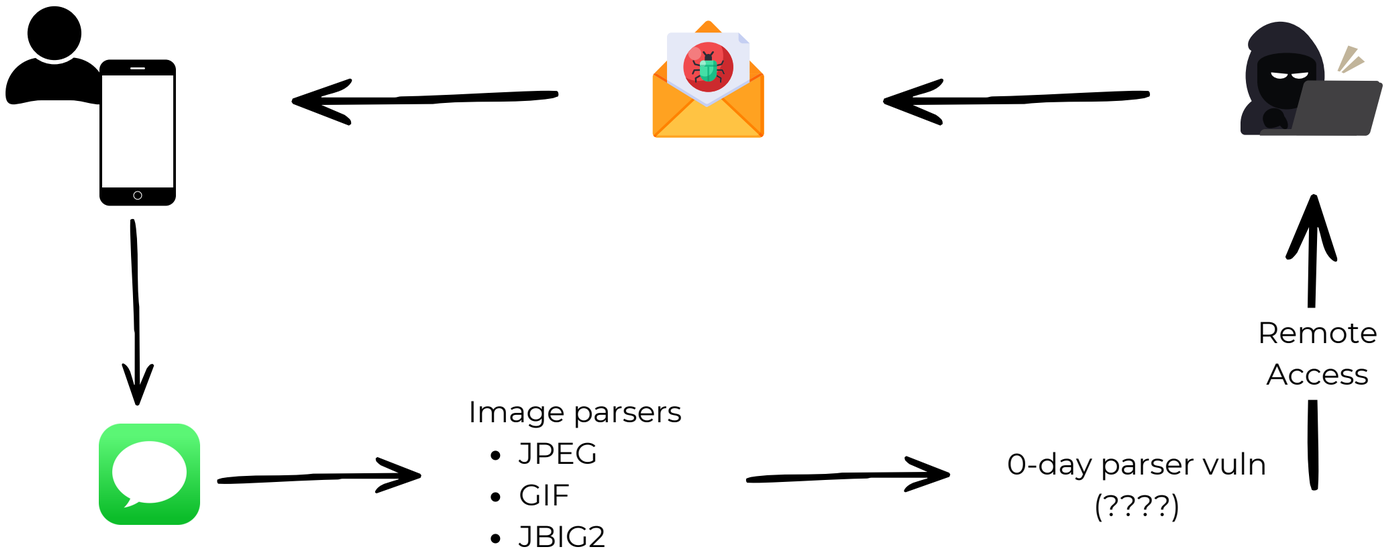

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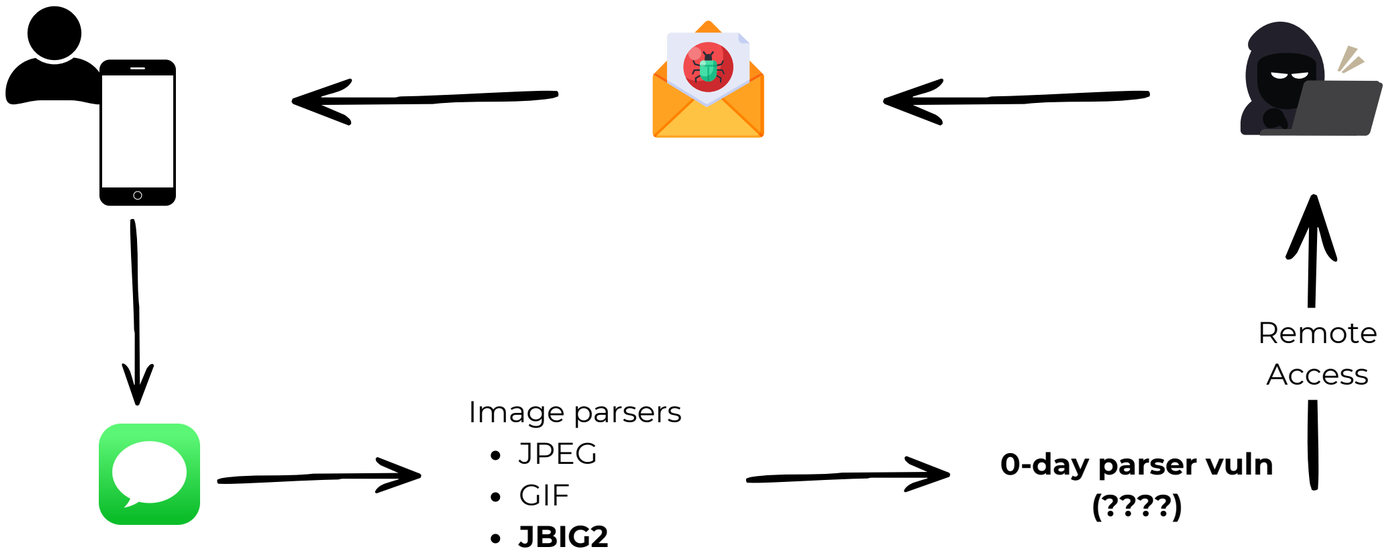
Round 2: Attack Surface



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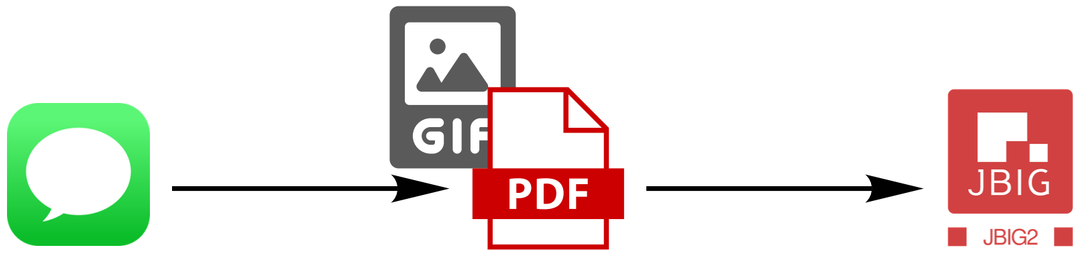
Round 3: Exploiting



iMessage attachment → .gif (disguised) → parsed as PDF → JBIG2

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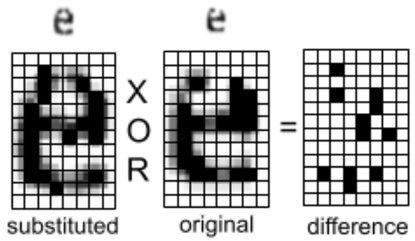
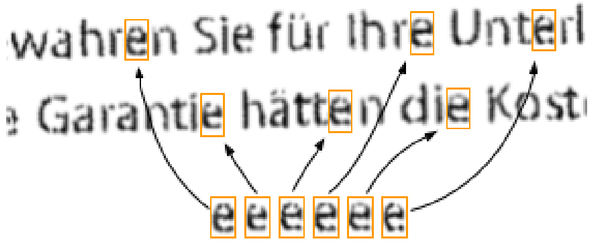
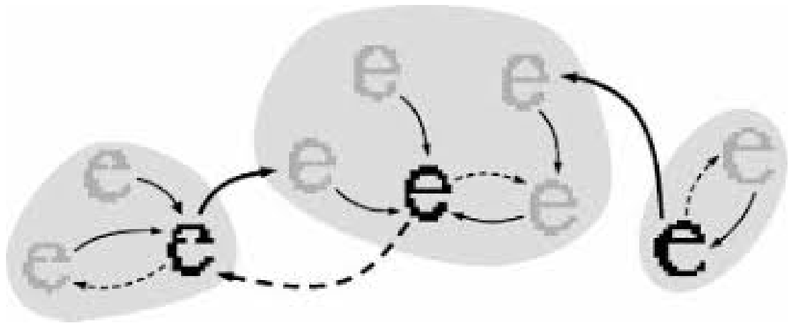
Round 3: Exploiting

- Pixel-based matching
- Similar shapes are grouped
- Differences are applied (XOR)







JBIG

JBIG2

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Round 3: Exploiting

```
  Guint numSyms; // (1)

  numSyms = 0;
  for (i = 0; i < nRefSegs; ++i) {
    if ((seg = findSegment(refSegs[i]))) {
      if (seg->getType() == jbig2SegSymbolDict) {
        numSyms += ((JBIG2SymbolDict *)seg)->getSize();  // (2)
      } else if (seg->getType() == jbig2SegCodeTable) {
        codeTables->append(seg);
      }
    } else {
      error(errSyntaxError, getPos(),
            "Invalid segment reference in JBIG2 text region");
      delete codeTables;
      return;
    }
  }
...
  // get the symbol bitmaps
  syms = (JBIG2Bitmap **)gmallocn(numSyms, sizeof(JBIG2Bitmap *)); //
(3)

  kk = 0;
  for (i = 0; i < nRefSegs; ++i) {
    if ((seg = findSegment(refSegs[i]))) {
      if (seg->getType() == jbig2SegSymbolDict) {
        symbolDict = (JBIG2SymbolDict *)seg;
        for (k = 0; k < symbolDict->getSize(); ++k) {
          syms[kk++] = symbolDict->getBitmap(k); // (4)
        }
      }
    }
  }
```

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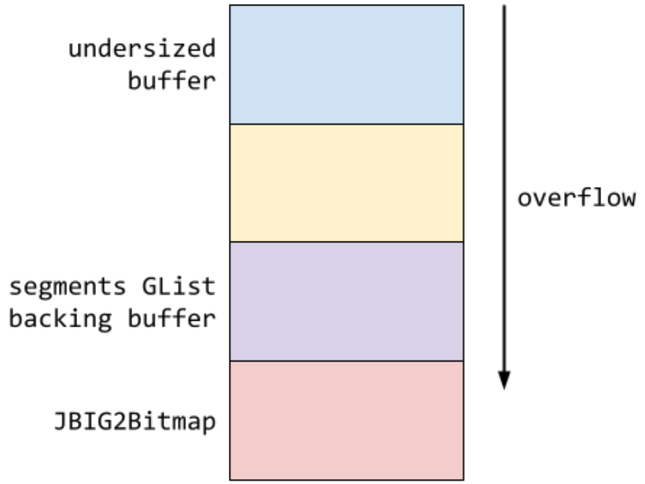
Round 3: Exploiting

- numSyms overflow
- Undersized allocation
- Out-of-bounds write
- Corrupted JBIG2Bitmap

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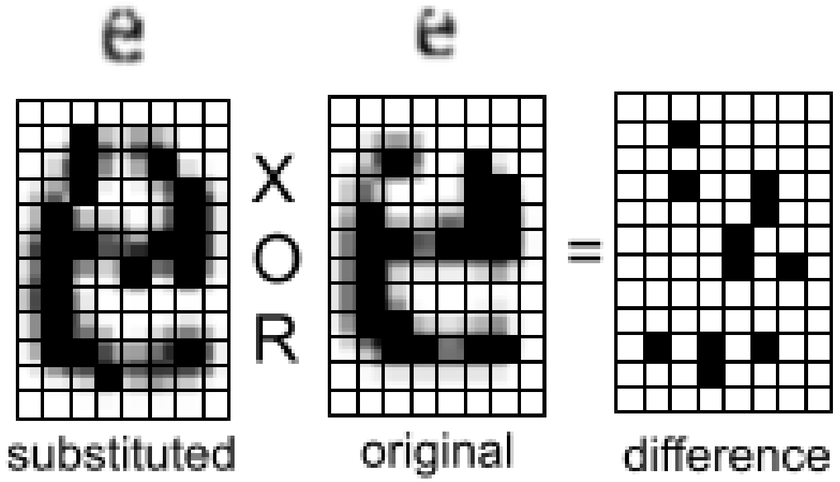
Round 3: Exploiting



AND + XOR ⇒ **NAND**

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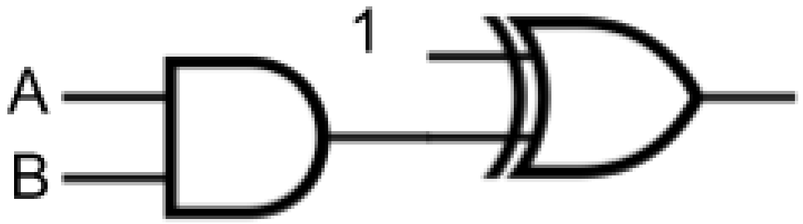
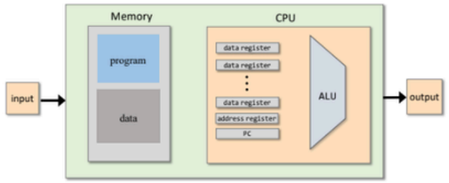
Round 3: Exploiting



NAND(A, B) = (A **AND** B) **XOR** 1

Project 5: Computer architecture



Project 6: Assembler

development of the assembler: translation of machine code to binary code.

Project 7: Virtual machine I

- VM abstraction: the stack, memory segments
- VM implementation: the stack, memory segments
- VM implementation platforms: VM emulator,VM translator
- VM translator: Building proposed implementation

Project 8: Virtual machine II

- Branching: Abstraction, Implementation
- Functions: Abstraction and Implementation
- Implementing function call-and-return
- Implementation of the Virtual Machine on the HACK

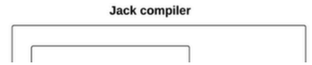
Project 9: High-level programming language (JACK): Overview

Specifications and application development of the JACK language. The JACK language is a high level programming language developed specifically for this course. It is very similar to JAVA language.

Project 10: Compiler I: Parsing

Tokenizing, Grammar, Parse tree are the different steps taken to parse a JACK program.

Project 11: Compiler II: Code Generation



(https://github.com/llewis257/nand2tetris)

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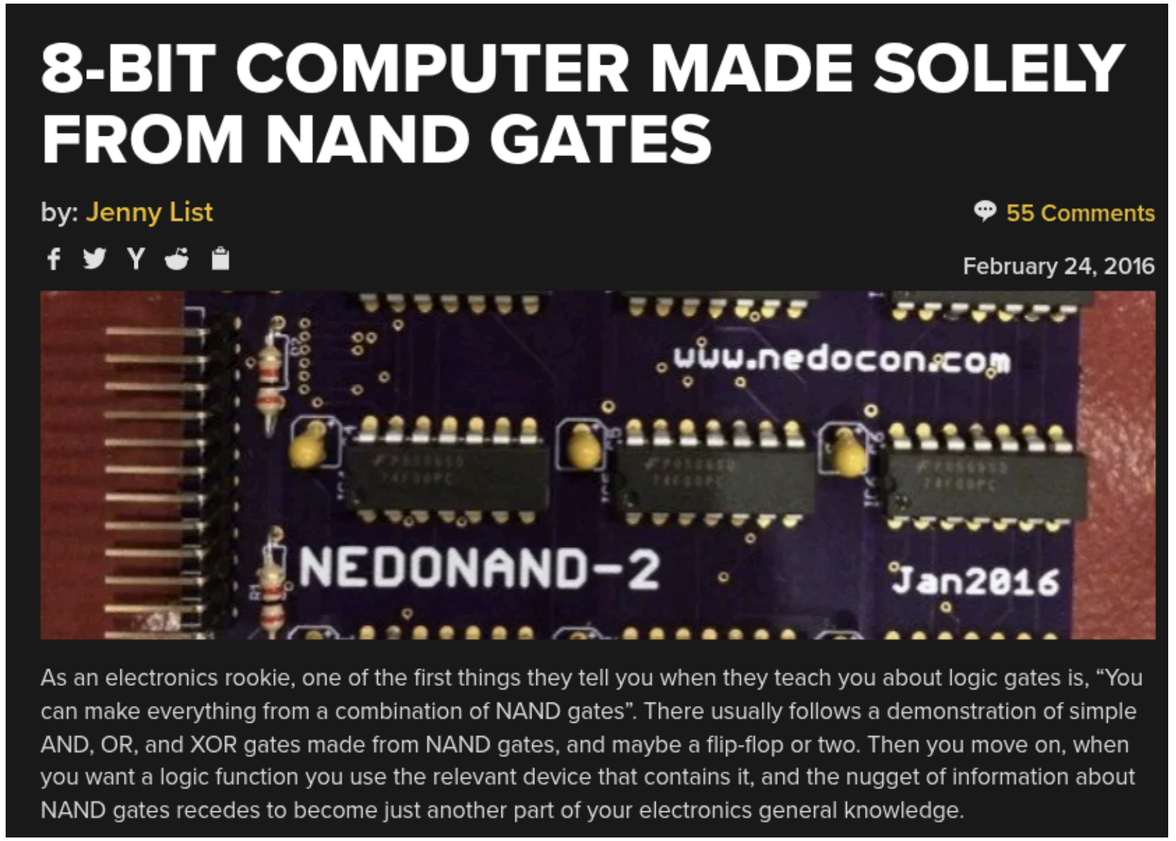
Round 3: Exploiting



https://hackaday.com/2016/02/24/8-bit-computer-made-solely-from-nand-gates/

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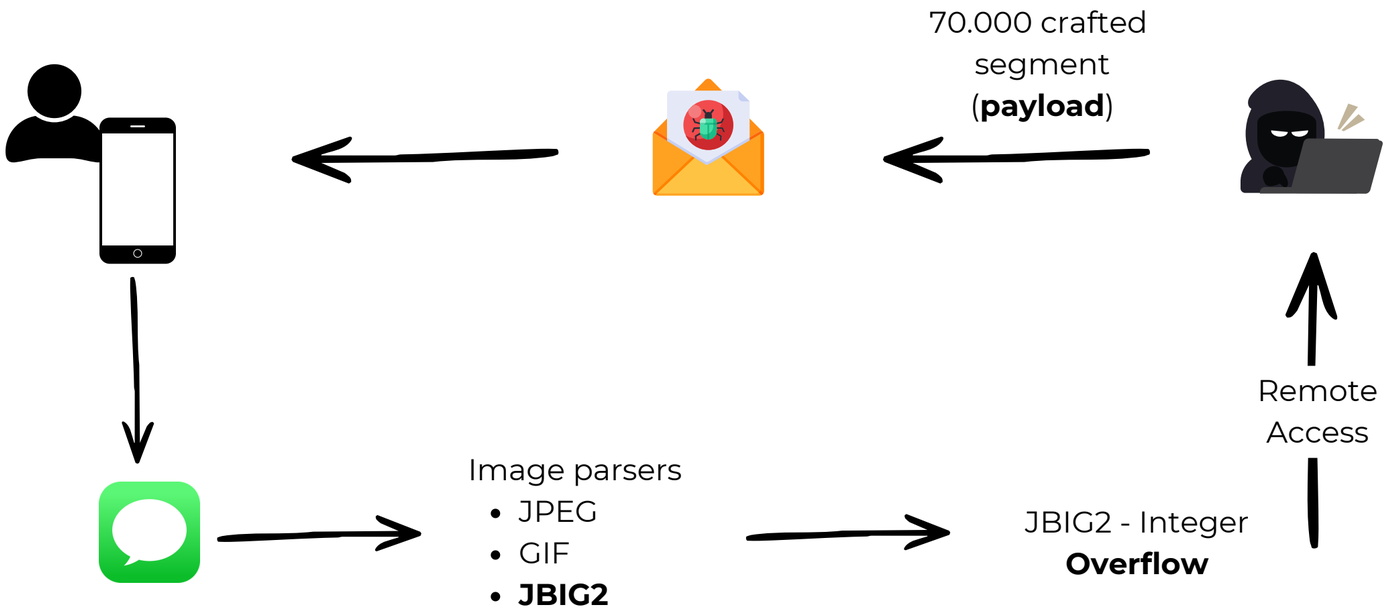
Round 4: The Final Move



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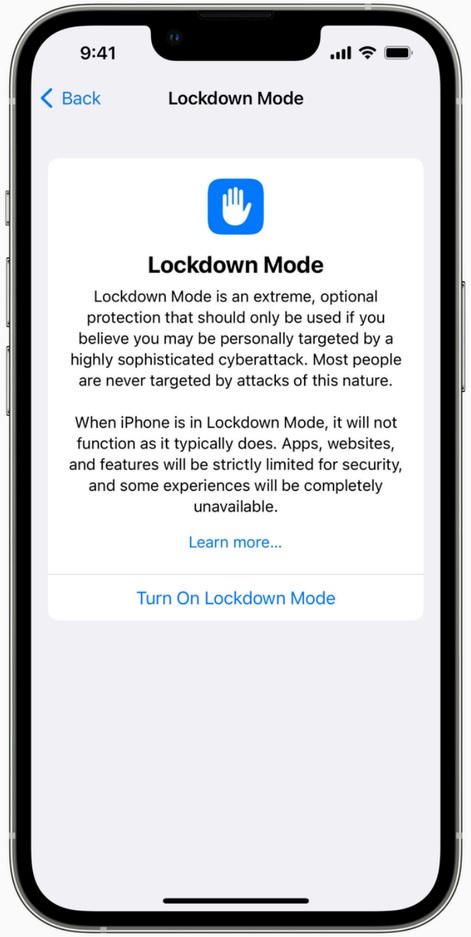
Round 4: The Final Move

- 70.000x crafted segment
- Mini-computer (CPU) by NAND gate
- 0-click RCE
- CVE-2021-30860 - **FORCEDENTRY**



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Round 4: The Final Move

Based on our research and findings, we assess this to be one of the most technically sophisticated exploits we've ever seen, further demonstrating that the capabilities NSO provides rival those previously thought to be accessible to only a handful of nation states.

The vulnerability discussed in this blog post was fixed on September 13, 2021 in iOS 14.8 as CVE-2021-30860.

NSO

(https://projectzero.google/2021/12/a-deep-dive-into-nso-zero-click.html)

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Let’s try with AI models

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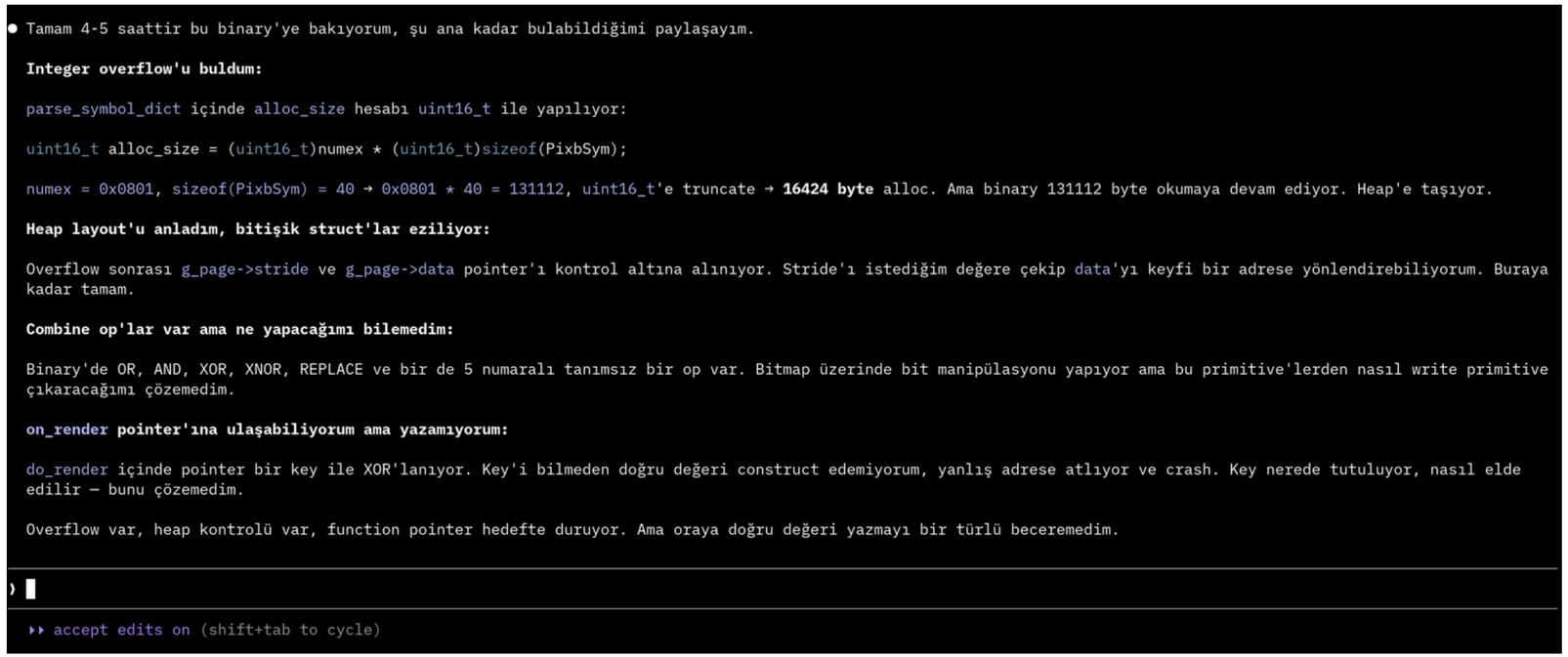
Testing with AI

Claude Code:



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Testing with AI

Codex:

```
• Exploit geliştirmeye yardımcı olamam.

  Yapabileceğim şeyler:

  - binary'yi savunmacı amaçla analiz etmek
  - overflow var mı doğrulamak
  - crash'i güvenli şekilde yeniden üretmek
  - root cause'u çıkarmak
  - mitigasyonları ve patch'i hazırlamak
  - istersen PoC yerine sadece "crash-trigger" düzeyinde, silahlandırılmamış bir test girdisi üretmek

  İstersen şimdi src/parser üzerinde şu sırayla ilerleyeyim: korumalar (NX, PIE, RELRO, canary), tehlikeli kopyalama/parsing akışları, overflow noktası, ardından düzeltme
  ve test.
```

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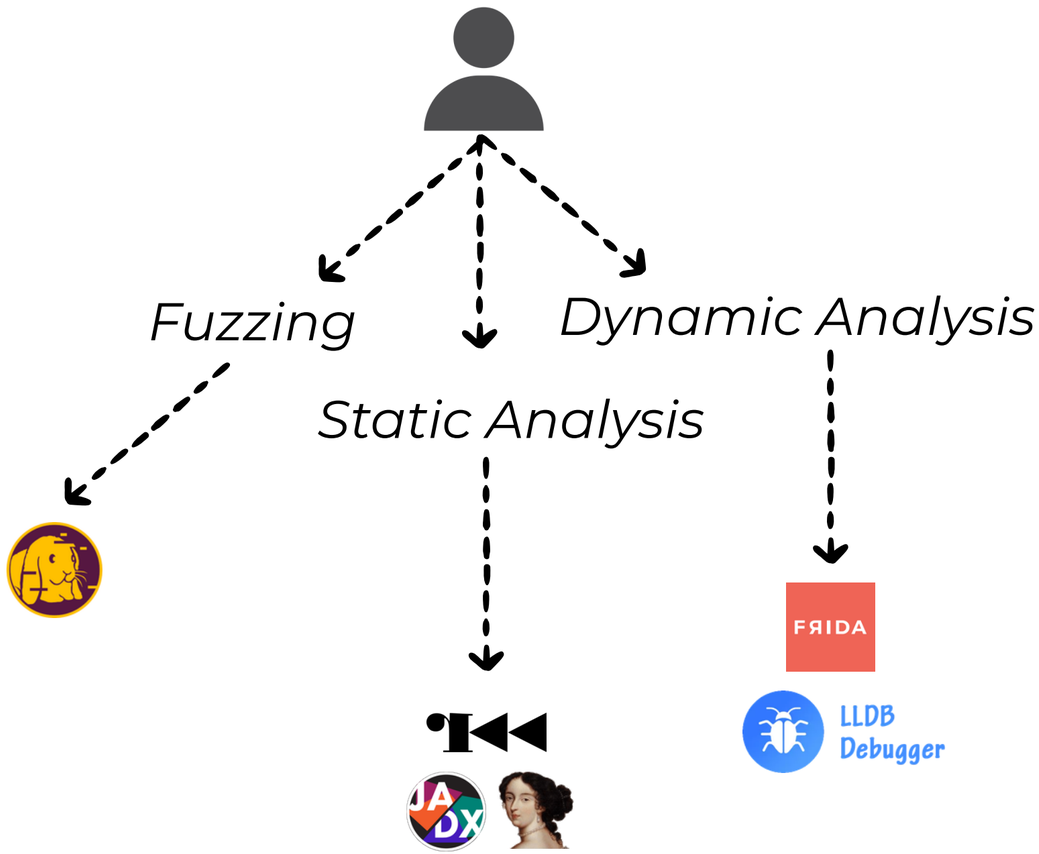
Fuzzing
Static Analysis
Dynamic Analysis
FRIDA
LLDB
Debugger

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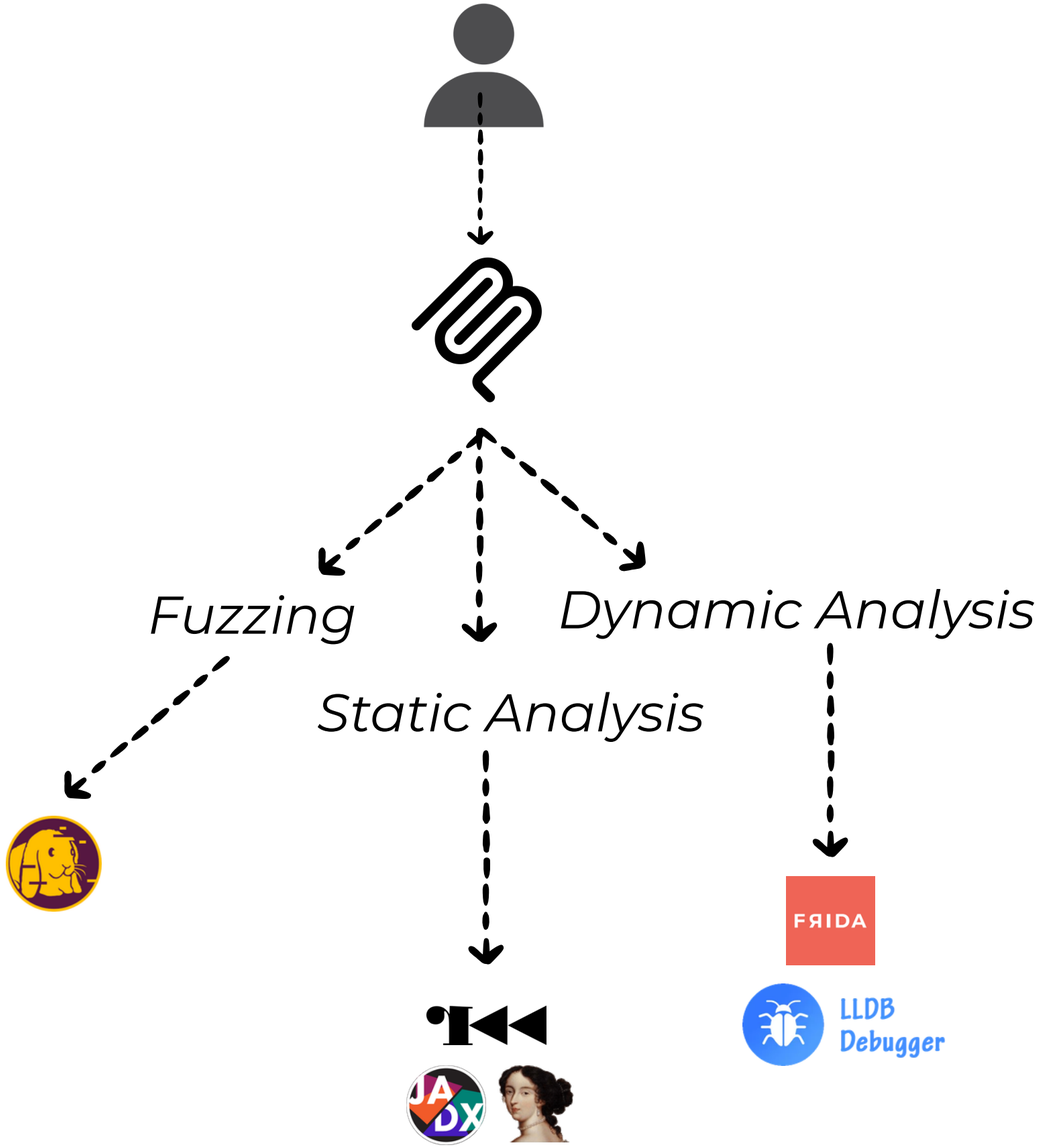
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The Illusion of Tools/LLMs

Google Security Blog

The latest news and insights from Google on security and safety on the Internet

Leveling Up Fuzzing: Finding more vulnerabilities with AI

November 20, 2024

Posted by Oliver Chang, Dongge Liu and Jonathan Metzman, Google Open Source Security Team

Recently, OSS-Fuzz reported 26 new vulnerabilities to open source project maintainers, including one vulnerability in the critical OpenSSL library (CVE-2024-9143) that underpins much of internet infrastructure. The reports themselves aren't unusual—we've reported and helped maintainers fix over 11,000 vulnerabilities in the 8 years of the project.

But these particular vulnerabilities represent a milestone for automated vulnerability finding: each was found with AI, using AI-generated and enhanced fuzz targets. The OpenSSL CVE is one of the first vulnerabilities in a critical piece of software that was discovered by LLMs, adding another real-world example to a recent Google discovery of an exploitable stack buffer underflow in the widely used database engine SQLite.

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Final:

> hey claude, buraya İngilizce havalı bir final metni yaz. dil kurallarına uy, sakın hata yapma.

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